

✓ DSE - Data Mining

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Assignment: data processing using a dataset

downloading ucimlrepo

since we will be using a health disease dataset from <https://archive.ics.uci.edu/dataset/45/heart+disease>

```
!pip install ucimlrepo
```

```
Requirement already satisfied: ucimlrepo in /usr/local/lib/python3.13/dist-packages (0.0.7)
Requirement already satisfied: pandas>=1.0.0 in /usr/local/lib/python3.13/dist-packages (from ucimlrepo)
Requirement already satisfied: certifi>=2020.12.5 in /usr/local/lib/python3.13/dist-packages (from pandas)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.13/dist-packages (from pandas)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.13/dist-packages (from pandas)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.13/dist-packages (from pandas)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.13/dist-packages (from pandas)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil)
```

importing the modules.

```
import pandas as pd
import numpy as np
from ucimlrepo import fetch_ucirepo
```

✓ importing the dataset from the site.

```
# fetch dataset
heart_disease = fetch_ucirepo(id=45)
```

```
# data (as pandas dataframes)
X = heart_disease.data.features
y = heart_disease.data.targets
df = pd.concat([X, y], axis=1)
```

```
# metadata
print(heart_disease.metadata)
# variable information
print(heart_disease.variables)
```

```
{'uci_id': 45, 'name': 'Heart Disease', 'repository_url': 'https://archive.ics.uci.edu/dataset/45/heart+disease'}
```

	name	role	type	demographic
0	age	Feature	Integer	Age
1	sex	Feature	Categorical	Sex
2	cp	Feature	Categorical	None
3	trestbps	Feature	Integer	None
4	chol	Feature	Integer	None
5	fb	Feature	Categorical	None
6	restecg	Feature	Categorical	None
7	thalach	Feature	Integer	None
8	exang	Feature	Categorical	None
9	oldpeak	Feature	Integer	None
10	slope	Feature	Categorical	None
11	ca	Feature	Integer	None
12	thal	Feature	Categorical	None
13	num	Target	Integer	None

	description	units	missing_values
0	None	years	no

1		None	None	no
2		None	None	no
3	resting blood pressure (on admission to the ho...	mm Hg		no
4	serum cholestoral	mg/dl		no
5	fasting blood sugar > 120 mg/dl	None		no
6		None	None	no
7	maximum heart rate achieved	None		no
8	exercise induced angina	None		no
9	ST depression induced by exercise relative to ...	None		no
10		None	None	no
11	number of major vessels (0-3) colored by flour...	None		yes
12		None	None	yes
13	diagnosis of heart disease	None		no

Dataset Description

```
print(df.info())
print(df.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 303 entries, 0 to 302
Data columns (total 14 columns):
```

#	Column	Non-Null Count	Dtype
0	age	303 non-null	int64
1	sex	303 non-null	int64
2	cp	303 non-null	int64
3	trestbps	303 non-null	int64
4	chol	303 non-null	int64
5	fbs	303 non-null	int64
6	restecg	303 non-null	int64
7	thalach	303 non-null	int64
8	exang	303 non-null	int64
9	oldpeak	303 non-null	float64
10	slope	303 non-null	int64
11	ca	299 non-null	float64
12	thal	301 non-null	float64
13	num	303 non-null	int64

```
dtypes: float64(3), int64(11)
```

```
memory usage: 33.3 KB
```

```
None
```

	age	sex	cp	trestbps	chol	fbs
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000
mean	54.438944	0.679868	3.158416	131.689769	246.693069	0.148515
std	9.038662	0.467299	0.960126	17.599748	51.776918	0.356198
min	29.000000	0.000000	1.000000	94.000000	126.000000	0.000000
25%	48.000000	0.000000	3.000000	120.000000	211.000000	0.000000
50%	56.000000	1.000000	3.000000	130.000000	241.000000	0.000000
75%	61.000000	1.000000	4.000000	140.000000	275.000000	0.000000
max	77.000000	1.000000	4.000000	200.000000	564.000000	1.000000

	restecg	thalach	exang	oldpeak	slope	ca
count	303.000000	303.000000	303.000000	303.000000	303.000000	299.000000
mean	0.990099	149.607261	0.326733	1.039604	1.600660	0.672241
std	0.994971	22.875003	0.469794	1.161075	0.616226	0.937438
min	0.000000	71.000000	0.000000	0.000000	1.000000	0.000000
25%	0.000000	133.500000	0.000000	0.000000	1.000000	0.000000
50%	1.000000	153.000000	0.000000	0.800000	2.000000	0.000000
75%	2.000000	166.000000	1.000000	1.600000	2.000000	1.000000
max	2.000000	202.000000	1.000000	6.200000	3.000000	3.000000

	thal	num
count	301.000000	303.000000
mean	4.734219	0.937294
std	1.939706	1.228536
min	3.000000	0.000000
25%	3.000000	0.000000
50%	3.000000	0.000000
75%	7.000000	2.000000
max	7.000000	4.000000

Cleaning and processing the data

```
#checking if any column has missing values
print(df.isnull().sum())
```

```
age          0
sex          0
cp           0
trestbps     0
chol         0
fbs          0
restecg      0
thalach      0
exang        0
oldpeak      0
slope        0
ca           4
thal         2
num          0
dtype: int64
```

- column `ca` and `thal` has missing values

we will fill `ca` (numerical) with median values, and `thal` (categorical) with mode values.

```
df['ca'] = df['ca'].fillna(df['ca'].median()) #median for ca
print(f"null values in ca is replaced with {df['ca'].median()}")
df['thal'] = df['thal'].fillna(df['thal'].mode()[0]) #mode for thal
print(f"null values in thal is replaced with {df['thal'].mode()[0]}")
```

```
null values in ca is replaced with 0.0
null values in thal is replaced with 3.0
```

```
print("null values in:")
print(f"ca: {df['ca'].isnull().sum()}")
print(f"thal: {df['thal'].isnull().sum()}")
```

```
null values in:
ca: 0
thal: 0
```

- Aggregation.

```
aggregation = df.groupby('num').agg({
    'age': 'mean',
    'trestbps': 'mean',
    'chol': 'mean',
    'thalach': 'mean'
})

print(aggregation)
```

	age	trestbps	chol	thalach
num				
0	52.585366	129.250000	242.640244	158.378049
1	55.381818	133.254545	249.109091	145.927273
2	58.027778	134.194444	259.277778	135.583333
3	56.000000	135.457143	246.457143	132.057143
4	59.692308	138.769231	253.384615	140.615385

```
aggregation = df.groupby('num').agg({
    'age': ['mean', 'min', 'max'],
    'chol': ['mean', 'median'],
    'trestbps': ['mean', 'max']
})
```

```
print(agggregation)
```

	age			chol		trestbps	
	mean	min	max	mean	median	mean	max
num							
0	52.585366	29	76	242.640244	234.5	129.250000	180
1	55.381818	35	70	249.109091	249.0	133.254545	192
2	58.027778	42	69	259.277778	254.0	134.194444	180
3	56.000000	39	70	246.457143	256.0	135.457143	200
4	59.692308	38	77	253.384615	231.0	138.769231	165

Discretization

```
chol_bins = [0, 200, 240, 1000]

chol_labels = [
    'Normal',
    'Borderline',
    'High'
]

df['chol_category'] = pd.cut(
    df['chol'],
    bins=chol_bins,
    labels=chol_labels
)

print(df[['chol', 'chol_category']].head())
```

	chol	chol_category
0	233	Borderline
1	286	High
2	229	Borderline
3	250	High
4	204	Borderline

Binarization

```
df['high_chol'] = (df['chol'] >= 240).astype(int)
df['high_bp'] = (df['trestbps'] >= 140).astype(int)
df['heart_disease'] = (df['num'] > 0).astype(int)
print(df[['chol', 'trestbps', 'high_chol', 'high_bp', 'heart_disease']].head())
```

	chol	trestbps	high_chol	high_bp	heart_disease
0	233	145	0	1	0
1	286	160	1	1	1
2	229	120	0	0	1
3	250	130	1	0	0
4	204	130	0	0	0

Sampling

selecting a subset of the entire dataset.

```
sample = df.sample(
    n=100,
    random_state=42
)

print(sample)
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	\
179	53	1	3	130	246	1	2	173	0	0.0	

228	54	1	4	110	206	0	2	108	1	0.0
111	56	1	4	125	249	1	2	144	1	1.2
246	58	1	4	100	234	0	0	156	0	0.1
60	51	0	4	130	305	0	0	142	1	1.2
..	...	...	..	...	...	...	...	...	...	...
163	58	0	4	100	248	0	2	122	0	1.0
155	70	1	4	130	322	0	2	109	0	2.4
97	60	0	4	150	258	0	2	157	0	2.6
68	59	1	4	170	326	0	2	140	1	3.4
229	66	1	4	112	212	0	2	132	1	0.1

	slope	ca	thal	num	chol_category	high_chol	high_bp	heart_disease
179	1	3.0	3.0	0	High	1	0	0
228	2	1.0	3.0	3	Borderline	0	0	1
111	2	1.0	3.0	1	High	1	0	1
246	1	1.0	7.0	2	Borderline	0	0	1
60	2	0.0	7.0	2	High	1	0	1
..	...	...	...	...	...	...	...	...
163	2	0.0	3.0	0	High	1	0	0
155	2	3.0	3.0	1	High	1	0	1
97	2	2.0	7.0	3	High	1	1	1
68	3	0.0	7.0	2	High	1	1	1
229	1	1.0	3.0	2	Borderline	0	0	1

[100 rows x 18 columns]